

Hence the total volume of the solid formed by the revolution of the area A about x -axis.

$$= \iint_A 2\pi y \, dx \, dy.$$

In polar coordinates, the above formula for the volume becomes

$$\iint_A 2\pi r \sin \theta \cdot r \, d\theta \, dr, \text{ i.e. } \iint_A 2\pi r^2 \sin \theta \, d\theta \, dr$$

Similarly, the volume of the solid formed by the revolution of the area A

$$\text{about } y\text{-axis} = \iint_A 2\pi x \, dx \, dy.$$

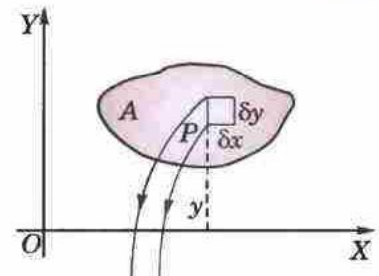


Fig. 7.25

Example 7.22. Calculate by double integration, the volume generated by the revolution of the cardioid $r = a(1 - \cos \theta)$ about its axis.

Solution. Required volume

$$= \int_0^\pi \int_0^{a(1-\cos \theta)} 2\pi r^2 \sin \theta \, dr \, d\theta$$

$$= 2\pi \int_0^\pi \left[\frac{r^3}{3} \right]_0^{a(1-\cos \theta)} \sin \theta \, d\theta$$

$$= \frac{2\pi a^3}{3} \int_0^\pi (1 - \cos \theta)^3 \cdot \sin \theta \, d\theta = \frac{2\pi a^3}{3} \left[\frac{(1 - \cos \theta)^4}{4} \right]_0^\pi = \frac{8\pi a^3}{3}.$$

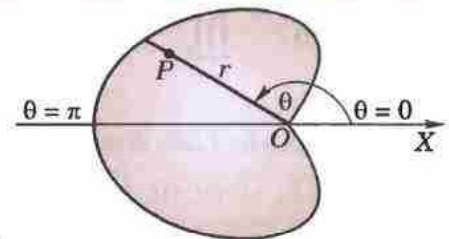


Fig. 7.26

7.7 CHANGE OF VARIABLES

An appropriate choice of co-ordinates quite often facilitates the evaluation of a double or a triple integral. By changing the variables, a given integral can be transformed into a simpler integral involving the new variables.

(1) In a double integral, let the variables x, y be changed to the new variables u, v by the transformation.

$$x = \phi(u, v), y = \psi(u, v)$$

where $\phi(u, v)$ and $\psi(u, v)$ are continuous and have continuous first order derivatives in some region R'_{uv} in the uv -plane which corresponds to the region R_{xy} in the xy -plane. Then

$$\iint_{R_{xy}} f(x, y) \, dx \, dy = \iint_{R'_{uv}} f[\phi(u, v), \psi(u, v)] |J| \, du \, dv \quad \dots(1)$$

where

$$J = \frac{\partial(x, y)}{\partial(u, v)} (\neq 0)$$

is the Jacobian of transformation* from (x, y) to (u, v) coordinates.

(2) For triple integrals, the formula corresponding to (1) is

$$\iiint_{R_{xyz}} f(x, y, z) \, dx \, dy \, dz = \iiint_{R'_{uvw}} f[x(u, v, w), y(u, v, w), z(u, v, w)] |J| \, du \, dv \, dw$$

where

$$J = \frac{\partial(x, y, z)}{\partial(u, v, w)} (\neq 0)$$

is the Jacobian of transformation from (x, y, z) to (u, v, w) coordinates.

Particular cases :

(i) To change cartesian coordinates (x, y) to polar coordinates (r, θ) , we have $x = r \cos \theta, y = r \sin \theta$ and

$$J = \frac{\partial(x, y)}{\partial(r, \theta)} = r$$

[Ex. 5.25, p. 216]

$$\therefore \iint_{R_{xy}} f(x, y) \, dx \, dy = \iint_{R'_{r\theta}} f(r \cos \theta, r \sin \theta) \cdot r \, dr \, d\theta.$$

* See footnote page 215.

(ii) To change rectangular coordinates (x, y, z) to cylindrical coordinates (ρ, ϕ, z) — Fig. 8.27, we have

$$x = \rho \cos \phi, y = \rho \sin \phi, z = z$$

and

$$J = \frac{\partial(x, y, z)}{\partial(\rho, \phi, z)} = \rho \quad [\text{Ex. 5.25}]$$

$$\text{Then } \iiint_{R_{xyz}} f(x, y, z) dx dy dz = \iiint_{R'_{\rho\phi z}} f(\rho \cos \phi, \rho \sin \phi, z) \cdot \rho d\rho d\phi dz.$$

(iii) To change rectangular coordinates (x, y, z) to spherical polar coordinates (r, θ, ϕ) —Fig. 8.28, we have

$$x = r \sin \theta \cos \phi, y = r \sin \theta \sin \phi, z = r \cos \theta$$

and

$$J = \frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = r^2 \sin \theta \quad [\text{Ex. 5.25}]$$

$$\text{Then } \iiint_{R_{xyz}} f(x, y, z) dx dy dz = \iiint_{R'_{r\theta\phi}} f(r \sin \theta \cos \phi, r \sin \theta \sin \phi, r \cos \theta) \cdot r^2 \sin \theta dr d\theta d\phi$$

Example 7.23. Evaluate $\iint_R (x+y)^2 dx dy$, where R is the parallelogram in the xy -plane with vertices $(1, 0), (3, 1), (2, 2), (0, 1)$ using the transformation $u = x + y$ and $v = x - 2y$. (U.P.T.U., 2004)

Solution. The region R , i.e., parallelogram $ABCD$ in the xy -plane becomes the region R' , i.e., rectangle $A'B'C'D'$ in the uv -plane as shown in Fig. 7.27, by taking

$$u = x + y \quad \text{and} \quad v = x - 2y \quad \dots(i)$$

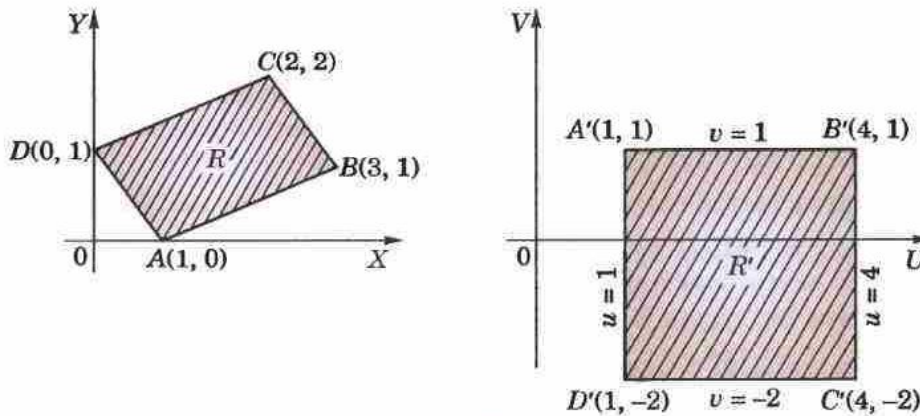


Fig. 7.27

From (i), we have

$$x = \frac{1}{3}(2u + v), y = \frac{1}{3}(u - v)$$

$\therefore$

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix} = -\frac{1}{3}$$

Hence, the given integral

$$= \iint_{R'} u^2 |J| du dv = \int_1^4 \int_{-2}^1 u^2 \cdot \frac{1}{3} \cdot du dv = \frac{1}{3} \left[\frac{u^3}{3} \right]_1^4 \cdot [v]_{-2}^1 = 21.$$

Example 7.24. Evaluate $\iint_D xy\sqrt{(1-x-y)} dx dy$ where D is the region bounded by $x = 0, y = 0$ and $x + y = 1$ using the transformation $x + y = u, y = uv$. (Marathwada, 2008)

Solution. We have $x = u - uv, y = uv$

$$\therefore J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \partial x / \partial u & \partial y / \partial u \\ \partial x / \partial v & \partial y / \partial v \end{vmatrix} = \begin{vmatrix} 1-v & v \\ -u & u \end{vmatrix} = u.$$

Also when $x = 0, u = 0, v = 1$; when $y = 0, u = 0, v = 0$ and when $x + y = 1, u = 1$

$\therefore$ the limits of u are from 0 to 1 and limits of v are from 0 to 1.

$$\begin{aligned} \text{Thus } \iint_D xy \sqrt{(1-x-y)} dx dy &= \int_0^1 \int_0^1 u(1-v) uv (1-u)^{1/2} |J| du dv \\ &= \int_0^1 \int_0^1 u^3 (1-u)^{1/2} v(1-v) du dv \\ &= \int_0^1 u^3 (1-u)^{1/2} du \times \int_0^1 v(1-v) dv \\ &= \int_0^{\pi/2} \sin^6 \theta \cos \theta \cdot 2 \sin \theta \cos \theta d\theta \times \left| \frac{v^2}{2} - \frac{v^3}{3} \right|_0^1 \\ &= 2 \int_0^{\pi/2} \sin^7 \theta \cos^2 \theta d\theta \left(\frac{1}{6} \right) = \frac{1}{3} \cdot \frac{6 \cdot 1}{9 \cdot 7 \cdot 5 \cdot 3} = \frac{2}{945}. \end{aligned}$$

$$\begin{aligned} \text{where } u &= \sin^2 \theta \\ du &= 2 \sin \theta \cos \theta d\theta \\ u = 0, \theta &= 0 \\ u = 1, \theta &= \pi/2 \end{aligned}$$

Example 7.25. Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by changing to polar coordinates. (Anna, 2003)

Hence show that $\int_0^\infty e^{-x^2} dx = \sqrt{\pi/2}$. (Madras, 2003; U.P.T.U., 2003; J.N.T.U., 2000)

Solution. The region of integration being the first quadrant of the xy -plane, r varies from 0 to ∞ and θ varies from 0 to $\pi/2$. Hence,

$$\begin{aligned} I &= \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy = \int_{\theta=0}^{\pi/2} \int_{r=0}^\infty e^{-r^2} r dr d\theta \\ &= -\frac{1}{2} \int_0^{\pi/2} \left\{ \int_0^\infty e^{-r^2} (-2r) dr \right\} d\theta = -\frac{1}{2} \int_0^{\pi/2} \left| e^{-r^2} \right|_0^\infty d\theta = \frac{1}{2} \int_0^{\pi/2} d\theta = \frac{\pi}{4}. \end{aligned} \quad \dots(i)$$

$$\text{Also } I = \int_0^\infty e^{-x^2} dx \times \int_0^\infty e^{-y^2} dy = \left\{ \int_0^\infty e^{-x^2} dx \right\}^2 \quad \dots(ii)$$

$$\text{Thus, from (i) and (ii), we have } \int_0^\infty e^{-x^2} dx = \sqrt{\pi/2}. \quad \dots(iii)$$

Example 7.26. Find the volume bounded by the paraboloid $x^2 + y^2 = az$, the cylinder $x^2 + y^2 = 2ay$ and the plane $z = 0$.

Solution. The required volume is found by integrating $z = (x^2 + y^2)/a$ over the circle $x^2 + y^2 = 2ay$.

Changing to polar coordinates in the xy -plane, we have $x = r \cos \theta$, $y = r \sin \theta$ so that $z = r^2/a$ and the polar equation of the circle is $r = 2a \sin \theta$.

To cover this circle, r varies from 0 to $2a \sin \theta$ and θ varies from 0 to π . (Fig. 7.28)

Hence the required volume

$$\begin{aligned} &= \int_0^\pi \int_0^{2a \sin \theta} z \cdot r d\theta dr = \frac{1}{a} \int_0^\pi d\theta \int_0^{2a \sin \theta} r^3 dr \\ &= \frac{1}{a} \int_0^\pi d\theta \left| \frac{r^4}{4} \right|_0^{2a \sin \theta} = 4a^3 \int_0^\pi \sin^4 \theta d\theta = \frac{3\pi a^3}{2}. \end{aligned}$$

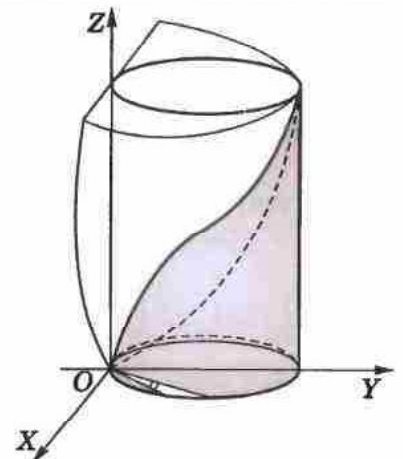


Fig. 7.28

Example 7.27. Find, by triple integration, the volume of the sphere $x^2 + y^2 + z^2 = a^2$.

(Bhopal, 2009 ; Madras, 2006 ; V.T.U., 2003 S)

Solution. Changing to polar spherical coordinates by putting

$$x = r \sin \theta \cos \phi, y = r \sin \theta \sin \phi, z = r \cos \theta$$

we have $dx dy dz = r^2 \sin \theta dr d\theta d\phi$.

Also the volume of the sphere is 8 times the volume of its portion in the positive octant for which r varies from 0 to a , θ varies from 0 to $\pi/2$ and ϕ varies from 0 to $\pi/2$.

$\therefore$ volume of the sphere

$$\begin{aligned} &= 8 \int_0^a \int_0^{\pi/2} \int_0^{\pi/2} r^2 \sin \theta dr d\theta d\phi = 8 \int_0^a r^2 dr \cdot \int_0^{\pi/2} \sin \theta d\theta \cdot \int_0^{\pi/2} d\phi \\ &= 8 \cdot \left[\frac{r^3}{3} \right]_0^a \cdot \left[-\cos \theta \right]_0^{\pi/2} \cdot \frac{\pi}{2} = 4\pi \cdot \frac{a^3}{3} \cdot (-0 + 1) = \frac{4}{3} \pi a^3. \end{aligned}$$

Example 7.28. Find the volume of the portion of the sphere $x^2 + y^2 + z^2 = a^2$ lying inside the cylinder $x^2 + y^2 = ay$. (Rohtak, 2003)

Solution. The required volume is easily found by changing to cylindrical coordinates (ρ, ϕ, z) . We therefore, have

$$x = \rho \cos \phi, y = \rho \sin \phi, z = z$$

and

$$J = \frac{\partial(x, y, z)}{\partial(\rho, \phi, z)} = \rho.$$

Then the equation of the sphere becomes $\rho^2 + z^2 = a^2$ and that of cylinder becomes $\rho = a \sin \phi$.

The volume inside the cylinder bounded by the sphere is twice the volume shown shaded in the Fig. 7.29 for which z varies from 0 to $\sqrt{a^2 - \rho^2}$, ρ varies from 0 to $a \sin \phi$ and ϕ varies from 0 to π .

$$\begin{aligned} \text{Hence the required volume} &= 2 \int_0^\pi \int_0^{a \sin \phi} \int_0^{\sqrt{a^2 - \rho^2}} \rho dz d\rho d\phi \\ &= 2 \int_0^\pi \int_0^{a \sin \phi} \rho \sqrt{a^2 - \rho^2} d\rho d\phi = 2 \int_0^\pi \left[-\frac{1}{3} (a^2 - \rho^2)^{3/2} \right]_0^{a \sin \phi} d\phi \\ &= \frac{2a^3}{3} \int_0^\pi (1 - \cos^3 \phi) d\phi = \frac{2a^3}{9} (3\pi - 4). \end{aligned}$$

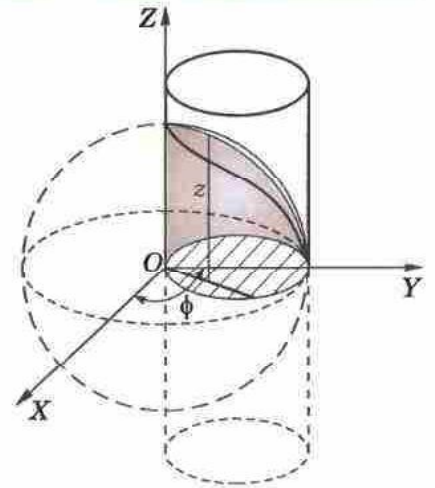


Fig. 7.29

Example 7.29. Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_{\sqrt{x^2+y^2}}^1 \frac{dz dy dx}{\sqrt{x^2+y^2+z^2}}$.

(V.T.U., 2008)

Solution. We change to spherical polar coordinates (r, θ, ϕ) , so that

$$x = r \sin \theta \cos \phi, y = r \sin \theta \sin \phi, z = r \cos \theta$$

and

$$J = r^2 \sin \theta, x^2 + y^2 + z^2 = r^2.$$

The region of integration is common to the cone $z^2 = x^2 + y^2$ and the cylinder $x^2 + y^2 = 1$ bounded by the plane $z = 1$ in the positive octant (Fig. 7.30). Hence θ varies from 0 to $\pi/4$, r varies from 0 to $\sec \theta$ and ϕ varies from 0 to $\pi/2$.

$\therefore$ given integral becomes

$$\begin{aligned} &\int_0^{\pi/2} \int_0^{\pi/4} \int_0^{\sec \theta} \frac{1}{r} \cdot r^2 \sin \theta dr d\theta d\phi = \int_0^{\pi/2} d\phi \int_0^{\pi/4} \left[\frac{r^2}{2} \right]_0^{\sec \theta} \sin \theta d\theta \\ &= \frac{\pi}{2} \int_0^{\pi/4} \frac{\sec^2 \theta}{2} \sin \theta d\theta = \frac{\pi}{4} \int_0^{\pi/4} \sec \theta \tan \theta d\theta = \frac{\pi}{4} \left[\sec \theta \right]_0^{\pi/4} = \frac{(\sqrt{2} - 1)\pi}{4}. \end{aligned}$$

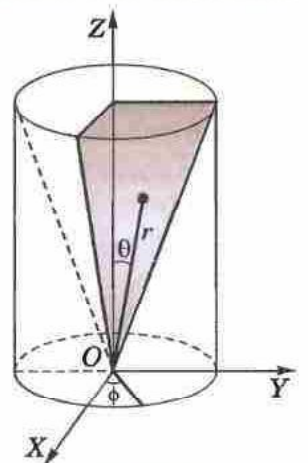


Fig. 7.30

Example 7.30. Find the volume of the solid surrounded by the surface

$$(x/a)^{2/3} + (y/b)^{2/3} + (z/c)^{2/3} = 1.$$

(Hissar, 2005 S)

Solution. Changing the variables, x, y, z to X, Y, Z where, $(x/a)^{1/3} = X, (y/b)^{1/3} = Y, (z/c)^{1/3} = Z$

i.e., $x = aX^3, y = bY^3, z = cZ^3$ so that $J = \partial(x, y, z)/\partial(X, Y, Z) = 27 abc X^2 Y^2 Z^2$.

$$\therefore \text{required volume} = \iiint dx dy dz = 27 abc \iiint X^2 Y^2 Z^2 dX dY dZ$$

taken throughout the sphere $X^2 + Y^2 + Z^2 = 1$.

...(i)

Now change X, Y, Z to spherical polar coordinates r, θ, ϕ so that $X = r \sin \theta \cos \phi, Y = r \sin \theta \sin \phi, Z = r \cos \theta$, and $\partial(X, Y, Z)/\partial(r, \theta, \phi) = r^2 \sin \theta$. To describe the positive octant of the sphere (i), r varies from 0 to 1, θ from 0 to $\pi/2$ and ϕ from 0 to $\pi/2$.

$$\begin{aligned} \therefore \text{required volume} &= 27abc \times 8 \int_0^1 \int_0^{\pi/2} \int_0^{\pi/2} r^2 \sin^2 \theta \cos^2 \phi \times r^2 \sin^2 \theta \sin^2 \phi \cdot r^2 \cos^2 \theta \cdot r^2 \sin \theta dr d\theta d\phi \\ &= 216 abc \int_0^1 r^8 dr \int_0^{\pi/2} \sin^4 \theta \cos^2 \theta d\theta \int_0^{\pi/2} \sin^2 \phi \cos^2 \phi d\phi = 4\pi abc/35. \end{aligned}$$

PROBLEMS 7.4

Evaluate the following integrals by changing to polar co-ordinates :

$$1. \int_0^1 \int_0^{\sqrt{1-y^2}} (x^2 + y^2) dy dx. \quad (\text{P.T.U., 2010}) \quad 2. \int_0^2 \int_0^{\sqrt{2x-x^2}} \frac{x dx dy}{x^2 + y^2} \quad (\text{Anna, 2009})$$

$$3. \int_0^{4a} \int_{y^2/4a}^y \frac{x^2 - y^2}{x^2 + y^2} dx dy \quad (\text{Mumbai, 2006})$$

$$4. \iint xy(x^2 + y^2)^{n/2} dx dy \text{ over the positive quadrant of } x^2 + y^2 = 4, \text{ supposing } n + 3 > 0. \quad (\text{S.V.T.U., 2007})$$

$$5. \iint \frac{dx dy}{(1 + x^2 + y^2)^2} \text{ over one loop of the lemniscate } (x^2 + y^2) = x^2 - y^2. \quad (\text{Mumbai, 2007})$$

$$6. \text{ Transform the following to cartesian form and hence evaluate } \int_0^\pi \int_0^a r^3 \sin \theta \cos \theta dr d\theta. \quad (\text{P.T.U., 2005})$$

$$7. \iint y^2 dx dy \text{ over the area outside } x^2 + y^2 - ax = 0 \text{ and inside } x^2 + y^2 - 2ax = 0. \quad (\text{Mumbai, 2006})$$

$$8. \text{ By using the transformation } x + y = u, y = uv, \text{ show that } \int_0^1 \int_0^{1-x} e^{y/(x+y)} dy dx = \frac{1}{2}(e - 1). \quad (\text{P.T.U., 2003})$$

$$9. \text{ Transform } \int_0^{\pi/2} \int_0^{\pi/2} \frac{\sqrt{\sin \phi}}{\sin \theta} d\phi d\theta \text{ by the substitution } x = \sin \phi \cos \theta, y = \sin \phi \sin \theta \text{ and show that its value is } \pi. \quad (\text{U.P.T.U., 2001})$$

Evaluate the following integrals by changing to spherical coordinates :

$$10. \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dx dy dz}{\sqrt{(1-x^2-y^2-z^2)}}. \quad (\text{V.T.U., 2006 ; Kottayam, 2005})$$

$$11. \iiint_V \frac{dx dy dz}{x^2 + y^2 + z^2} \text{ where } V \text{ is the volume of the sphere } x^2 + y^2 + z^2 = a^2. \quad (\text{Anna, 2009})$$

$$12. \text{ Evaluate } \iiint \frac{dx dy dz}{(1 + x + y + z)^3} \text{ over the volume of the tetrahedron } x = 0, y = 0, z = 0, x + y + z = 1. \quad (\text{Mumbai, 2007})$$

$$13. \text{ Show that } \iiint \frac{dx dy dz}{\sqrt{(a^2 - x^2 - y^2 - z^2)}} = \frac{\pi^2 a^3}{8}, \text{ the integral being extended for all the values of the variables for which the expression is real.} \quad (\text{U.T.U., 2010})$$

$$14. \iiint z^2 dx dy dz, \text{ taken over the volume bounded by the surfaces } x^2 + y^2 = a^2, x^2 + y^2 = z \text{ and } z = 0.$$

15. Find the volume bounded by the xy -plane, the cylinder $x^2 + y^2 = 1$ and the plane $x + y + z = 3$. (I.S.M., 2001)
16. Find the volume bounded by the xy -plane, the paraboloid $2z = x^2 + y^2$ and the cylinder $x^2 + y^2 = 4$. (Raipur, 2005)
17. Find the volume cut from the sphere $x^2 + y^2 + z^2 = a^2$ by the cone $x^2 + y^2 = z^2$. (S.V.T.U., 2006)
18. Find the volume common to the cylinders $x^2 + y^2 = a^2$ and $x^2 + z^2 = a^2$. (U.P.T.U., 2006)
19. Find the volume cut off from the cylinder $x^2 + y^2 = ax$ by the planes $z = 0$ and $z = x$. (Marathwada, 2008)
20. Find the volume enclosed by the cylinders $x^2 + y^2 = 2ax$ and $z^2 = 2ax$. (Marathwada, 2008)
21. Find the volume of the cylinder $x^2 + y^2 - 2ax = 0$, intercepted between the paraboloid $x^2 + y^2 = 2az$ and the xy -plane.
22. Find the volume bounded by the cylinder $x^2 + y^2 = 4$ and the hyperboloid $x^2 + y^2 - z^2 = 1$.
23. Find the volume of the region bounded by $z = x^2 + y^2$, $z = 0$, $x = -a$, $x = a$ and $y = -a$, $y = a$.
24. Prove, by using a double integral that the volume generated by the revolution of the cardioid $r = a(1 + \cos \theta)$ about its axis is $8\pi a^3/3$. (V.T.U., 2000)
25. Evaluate $\iiint (x + y + z) dx dy dz$ over the tetrahedron bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $x + y + z = 1$. [See Fig. 7.34]
26. Find the volume of the tetrahedron bounded by the coordinate planes and the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. (Burdwan, 2003)
27. Work out example 7.21 by changing the variables.

7.8 AREA OF A CURVED SURFACE

Consider a point P of the surface $S : z = f(x, y)$. Let its projection on the xy -plane be the region A . Divide it into area elements by drawing lines parallel to the axes of X and Y . (Fig. 7.31).

On the element $\delta x \delta y$ as base, erect a cylinder having generators parallel to OZ and meeting the surface S in an element of area δS .

As $\delta x \delta y$ is the projection of δS on the xy -plane,

$\therefore \delta x \delta y = \delta S \cdot \cos \gamma$, where γ is the angle between the xy -plane and the tangent plane to S at P , i.e., it is the angle between the Z -axis and the normal to S at $P (= \angle Z'PN)$.

Now since the direction cosines of the normal to the surface $F(x, y, z) = 0$ proportional to

$$\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z}.$$

$\therefore$ the direction cosines of the normal to $S [F = f(x, y) - z]$ are proportional to $-\frac{\partial z}{\partial x}, -\frac{\partial z}{\partial y}, 1$ and those of the z -axis are $0, 0, 1$.

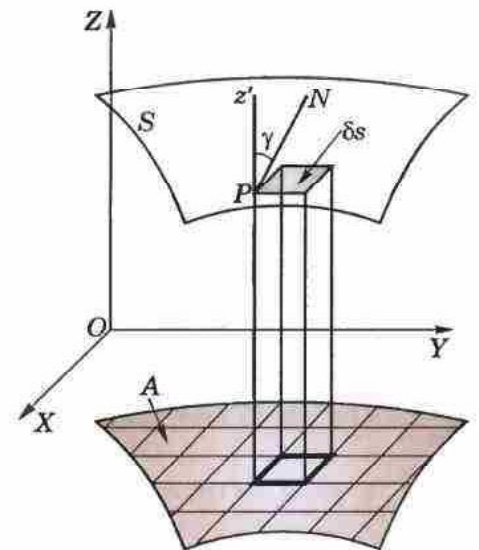


Fig. 7.31

$$\text{Hence } \cos \gamma = \frac{1}{\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}} \quad \therefore \delta S = \frac{\delta x \delta y}{\cos \gamma} = \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} \delta x \delta y$$

$$\text{Hence } S = \lim_{\delta S \rightarrow 0} \sum \delta S = \iint_A \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} dx dy$$

Similarly, if B and C be the projections of S on the yz - and zx -planes respectively, then

$$S = \iint_B \sqrt{\left(\frac{\partial x}{\partial y}\right)^2 + \left(\frac{\partial x}{\partial z}\right)^2 + 1} dy dz$$

and

$$S = \iint_C \sqrt{\left(\frac{\partial y}{\partial z}\right)^2 + \left(\frac{\partial y}{\partial x}\right)^2 + 1} dz dx.$$